Muminul Islam

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Research Interests

Reliable LLMs and vision-language models; model evaluation and uncertainty estimation; multilingual and low-resource NLP; knowledge distillation and efficient model adaptation; trustworthy retrieval-augmented systems.

Education

United International University 2020 – 2025
B.Sc. in Computer Science and Engineering *Dhaka, Bangladesh*

- **CGPA:** 3.00/4.00(approx 80%).
- **Undergraduate thesis:** Developed a Bangla–Chittagonian parallel corpus of 5,000+ sentence pairs and worked with Transformer-based translation models; related dataset manuscript completed its third revision for *Data in Brief* (decision pending).

Research

RE-SciVQA: Scientific Visual Question Answering Nov. 2025 – Sep. 2026
Academic Research Collaborator — Dr. Md Adnan Arefeen, NSU *[arXiv]*

- Investigated evaluation errors in scientific chart QA, developing range- and tolerance-aware scoring and confidence-gated answer verification.
- Contributed to QLoRA adaptation of a 4B vision-language model and implemented context-aware retrieval with query expansion.
- *Publication status:* 2026 arXiv preprint; revised manuscript in preparation.

Grounded Multimodal Reasoning in Dermatology Sep. 2026 – Present
ELITE Research Lab *Manuscript in preparation*

- Systematically reviewed 96 source records, consolidated into 75 distinct title-level records, examining visual grounding, clinical evidence, and diagnostic reliability.
- Contributed to an evaluation framework addressing grounding faithfulness, uncertainty, skin-tone fairness, and robustness in clinical multimodal AI.

ReflectBeforeAI: AI-Assisted Student Reasoning 2026 – Present
HCI Design Research Project — BRAC University *In progress*

- An interactive prototype and evaluation plan to study how reflection prompts and optional verification influence independent reasoning and reliance on AI.

Professional Experience

Explainable LLM and Interpretable Technology Ensemble Research Lab September 2026 – Present
Researcher (Part-Time) *Remote – Queens, NY, USA*

Harris Digital (Harris Web Works) June 2025 – November 2025
Trainee Software Engineer *Dhaka, Bangladesh*

- Introduced AI-assisted development and prompt-engineering workflows using tools such as ChatGPT, OpenAI Codex, and GitHub Copilot to accelerate prototyping and development.

Competitive Programming & Achievements

- **Champion** — UIU Intra-University AI Contest, Fall 2022; developed a multiclass classification model to predict medical conditions from research-paper abstracts.
- **ICPC Asia Dhaka Regional 2024**: Rank 135/309 onsite and Rank 172/2400+ in the online preliminary; Honourable Mention in 2021, 2022, and 2023.
- **Inter-University Programming Contests**: Rank 7 (UAP Collaborative Contest 2024), Rank 8 (UIU Coders Combat 3.0), Rank 50 (KUET IUPC 2025), Rank 52 (SUST IUPC 2024), and Rank 53 (IUT IUPC 2024).
- Solved **800+ algorithmic problems** across **150+ contests** on Codeforces (B4TMAN, max rating 1332), VJudge, and other online judges.
- Awarded the **UIU Merit Scholarship twice** for academic performance and placed **3rd Runner-up** at the System Analysis and Design Lab Project Show.

Technical Skills

- **Languages:** Python, C, C++, C#, TypeScript. **ML:** PyTorch, Transformers, scikit-learn, TensorFlow, NumPy, Pandas, QLoRA, RAG.
- **Software and tools:** Git, GitHub, Linux, Bash, REST APIs, ASP.NET Core, React, PostgreSQL, MySQL.

Selected Projects

- **E-Commerce Inventory API** | *ASP.NET Core, EF Core, JWT, PostgreSQL*
  Developed a secure .NET 8 REST API with JWT/BCrypt authentication, product/category CRUD, pagination, search, soft deletes, repository-pattern data access, and PostgreSQL support. [GitHub]
- **Codewar** | *PHP, MySQL, Bootstrap*
  Developed a competitive-programming platform with practice problems and analytics dashboards. [GitHub]
- **LinguaLearner** | *React Native, Express.js, TypeScript, MongoDB*
  Built a cross-platform language-learning application with quizzes, blogs, and secure user authentication. [GitHub]
- **Cinephiles** | *PHP, MySQL, JavaScript*
  Developed a movie platform featuring personalised recommendations, watch-party rooms, and community interaction. [GitHub]

English Language Proficiency

IELTS Academic: Overall 7.0 (Listening 8.0, Reading 7.5, Writing 6.5, Speaking 6.5).

Academic References

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